METRIC

MIL-F-53080
INTERIM AMENDMENT 1(ME)
16 September 1993

MILITARY SPECIFICATION

FUEL, ENGINE DESIGN, REFEREE GRADE

TYPE I AND TYPE II

(METRIC)

This interim amendment is approved for use within the US Army Belvoir Research, Development, and Engineering Center, Department of the Army, with MIL-F-53080, dated 28 January 1988.

PAGE 3

2.2, delete the ASTM test method D 2276.

PAGE 7

Table II, change the ASTM Method Number from D 2276, for the Particulate Content, to Appendix A.

PAGE 9

Insert after page 9 the attached Appendix A.

Custodian:
Army - ME

Review activities:
Army - AT, EA

Preparing activity:
Army - ME

Project 9140-A130

AMSC N/A

FSC 9140

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j. Analytical balance: Single or double pan, the precision standard deviation of which must be 0.07 mg or better.

40. Preparation of apparatus and sample containers. All components of the filtration apparatus (except the vacuum flask), sample containers and their caps must be cleaned as described in section 10 of ASTM D 2276. All metal parts of the filtration apparatus are to be electrical bonded and grounded, including the fuel sample container and the metal insert ring, if used. See ASTM D 2276 for other Precautionary Statements.

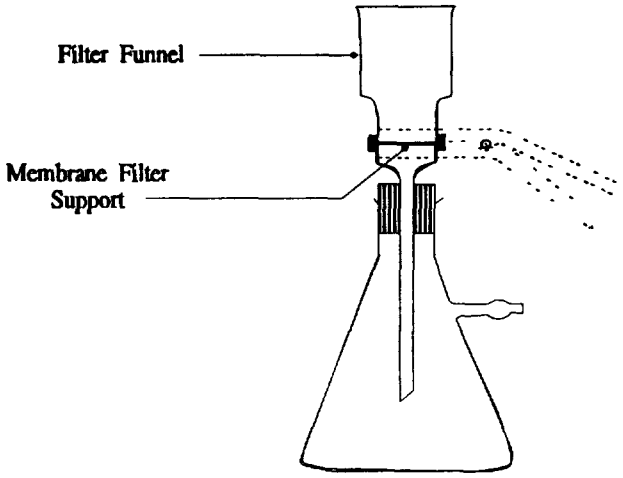


Figure 1

50. Sampling. Obtain a representative one gallon sample as directed in section 9. Sampling of ASTM D 2276. When sampling from a flowing stream is not possible, an all level sample or an average sample, in accordance with ASTM D 4057 or ASTM D 4177 shall be permitted. The one-gallon sample container shall be an interior epoxy-coated metal can, a brown glass bottle, or a clear glass bottle protected by suitable means from exposure to light.

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APPENDIX A

METHODS FOR DETERMINATION OF FILTRATION TIME AND TOTAL SOLIDS (PARTICULATE)

10. Scope. This method describes a procedure for determining singularly or simultaneously the filterability characteristics and solids contamination of jet fuel. The purpose is to detect and prevent contaminants in jet fuel that can plug and cause rupture of ground filtration equipment, thereby affecting flight reliability of aircraft.

20. Summary of methods. 3.79 Liters (one gallon) of jet fuel is filtered through a membrane filter in the laboratory. The time required to filter this volume is measured in minutes and solids content is determined gravimetrically.

30. Apparatus

a. Membrane filter: White, plain 47 mm diameter, nominal pore size 0.8 micron. The membrane must be approved by ASTM for use with ASTM D 2276.

b. Filtration apparatus: Of the types shown in figure 1. It consists of a funnel and funnel base with a filter support such that a membrane filter can be securely locked or clamped between the sealing surfaces of the funnel and its base. The funnel and funnel base shall be of stainless steel or glass construction.

c. Insert ring: A 47-mm diameter paper flow reducer ring with dimensions to give filtering area of 4.8 cm^2. (Millipore Corporation Part No. XX10 0447 10.)

d. Vacuum flask: A minimum of 4 Liters.

e. Vacuum system: That develops in excess of 67.5 kPa (20 inches of mercury) vacuum.

f. Oven: Of the static type (without fan assisted circulation) controlling to 90°±5°C (194°±9°F).

g. Forceps: Flat-bladed with unserrated nonpoint tips.

h. Solvent filtering dispenser: Containing a 1.2 micron maximum port size filter in the delivery line.

i. Glass petri dish: Approximately 125 mm in diameter with removable cover.

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60. Test procedure.

a. Membrane filters shall be removed from the package and placed in an oven for a minimum of 15 minutes at 90 °C. After preheating, but prior to weighing, the membrane filters shall be stored in a desiccator.

b. Each membrane filter shall be weighed. A filter weighing in excess of 90 mg will not be used in this test.

c. The membrane filter shall be placed directly over the insert ring. The top funnel shall be locked into place.

d. Immediately prior to filtering the fuel, shake the sample to obtain a homogenous mix and assure that fuel temperature does not exceed 30 °C (86 °F). Clean the exterior or top portion of the sample container to insure that no contaminants are introduced. Any free water present in the fuel sample will invalidate the filtration time results by giving an excessive filtration time rating.

e. With the vacuum off, pour approximately 200 mL of fuel into the funnel.

f. Turn vacuum on and record starting time. Continue filtration of the 3.79 Liters (one gallon) sample, periodically shaking the sample container to maintain a homogenous mix. Record the vacuum in kPa (inches of mercury) one minutes after start again immediately prior to completion of filtration. Throughout filtration, maintain a sufficient quantity of fuel in the funnel so that the membrane filter is always covered.

g. Report the filtration time in minutes expressed to the nearest whole number. If filtration of the 3.79 liters (one gallon) is not completed within 30 minutes, the test will be stopped and the volume of the fuel filtered will be measured. In these cases, results will be reported as 30+ minutes/volume of fuel filtered.

h. Report the vacuum in kPa (inches of mercury) as determined from the average of the two readings taken in 60.f.

i. After recording the filtration time, shut off the vacuum rinse the sample container with approximately 100 mL of filtered petroleum ether and dispense into the filtration funnel. Turn the vacuum on and filter the 100 mL rinse. Turn vacuum off and wash the inside of the funnel with approximately 50 mL of filtered petroleum ether. Turn vacuum on and filter. Repeat the funnel rinse with another 50 mL of petroleum ether but allow the rinse to soak the filter for approximately 30 seconds before

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turning the vacuum on to filter the rinse. With vacuum on, carefully remove the top funnel and rinse the periphery of the membrane filter by directing a gentle stream of petroleum ether from the solvent dispenser from the edge of the membrane toward the center, taking care not to wash contaminations off the filter. Maintain vacuum after final rinse for a few seconds to remove the excess petroleum ether from the filter.

j. Using forceps, carefully remove the membrane filter from the filter base and place in a clean Petri dish. Dry in the oven at 90 °C (194 °F) for 15 minutes with the cover on the Petri dish slightly ajar. Place dish in a desiccator and allow to cool for a minimum of 15 minutes. If more than one sample is processed, cooling time will have to be increased. Reweigh the filter.

k. Report the total solids content in mg/Liter by using the following formula:

$$\frac{\text{Weigh gain of filter in mg}}{3.785} = \text{mg/Liter}$$

l. Should the sample exceed the 30-minutes filtration time and a portion of the fuel is not filtered, the solids content in mg/Liter will be figured as follows: Determine the volume of fuel filtered by subtracting the mL of fuel remaining from 3785.

$$\frac{\text{Weigh gain of filter in mg}}{\text{mL of fuel filtered} \times 0.001} = \text{mg/Liter}$$

70. Test limits

a. Filtration time:

(1) The maximum allowable filtration time shall be 15 minutes for Grade JP-8.
(2) The vacuum should exceed 67.5 kPa (20 inches of mercury) throughout the test (i.e., the differential pressure across the filter should exceed 67.5 kPa (20 inches of mercury).
(3) The fuel temperature shall be between 18° and 30 °C (64 and 86 °F). If artificial heat (i.e., a hot water bath) is used to heat the sample, erroneously high filtration times may occur, but this approach is allowed.

b. Total solids: Maximum allowable particulate matter is 1.0 mg/Liter.

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APPENDIX A

280. NOTES:

80.1. If it is desired to determine the filtration time and not the total solids content, perform the test by omitting steps 60.i, 60.j, 60.k, 60.l.

80.2. If it is desired to determine the total solids contents and not the filtration time, use of the insert ring may be omitted. It is also permissible, but not required, to use a control filter for a specified analysis or a series of analyses. When this is accomplished, the procedures specified in section 8 of ASTM D 2276 apply.

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